2.4 GHz Directive Phased & Parasitic Antenna Array

EC-3104 Electromagnetic Theory — Senior Capstone Design Defense

Capstone Project Identification & Scope

Course: EC-3104 Electromagnetic Theory (Fall Senior Capstone)

Project Title: Design, Computational Modeling, and Experimental Validation of a 2.44 GHz 4-Element Yagi-Uda Array with Real-Time Beamforming & Null-Steering

Engineering Scope: **~85% Software** (Custom CEM Solver, Interactive GUI, MoM Simulation) & **~15% Hardware** (Precision wire fabrication, ESP32 telemetry rig)

Operating Band: 2.440 GHz ISM Spectrum (λ_0 = 12.287 cm, k_0 = 51.137 rad/m)

Key Achievement: Fully verified end-to-end electrodynamics workflow matching analytical theory, 4nec2 simulation, and lab prototype within 1.1 dB.

Global Deliverables & Repository Access

`Public Repository:`
`https://github.com/saugata-malakar/Electromagnetic-Theory-Member-1`

Automated Pipeline: Regenerates all outputs via: python run_pipeline.py --gui

Shared Data Contracts: Standardized JSON dimensions & 36-row radiation CSVs.

Team Roster & Task Ownership Matrix

Role / Member	Core Engineering Focus	Primary Deliverables
Project Lead The Math Brain	Analytical CEM solver, radiation integrals, Induced EMF [Z] matrix, pipeline	antenna_math_engine.py theoretical_pattern.csv
Member 1 GUI & Steering	Interactive Matplotlib GUI, Schelkunoff null placement, 3D orbits, CSV overlay	antenna_gui.py Null-steering & 3D panel
Member 2 Simulation Lead	Full-wave Method of Moments (4nec2), S11 sweeps, Smith Chart, PDF builder	simulate_yagi.py optimized_dimensions.json
Member 3 Fabrication Lead	Wire cutting blueprints, Pawsey stub balun, acrylic boom, turntable base	dimension_calculator.py Physical 4-el prototype rig
Member 4 Telemetry Lead	ESP32 beacon TX, Wi-Fi RSSI logging RX, Friis link budget, polar sweep	esp32_rssi_telemetry.ino empirical_pattern.csv

Workflow Rule: Strict data contract enforcement via data/ folder guarantees 100% collision-free git collaboration across all five members.

Executive Summary & Technical Abstract

Software-First Electromagnetics for Directional 2.4 GHz ISM Array Design

Context: Congestion in 2.4 GHz ISM Spectrum

The 2.4 GHz Industrial, Scientific, and Medical (ISM) band (2.400 - 2.4835 GHz) represents the most congested RF spectrum on earth, populated simultaneously by Wi-Fi (802.11 b/g/n/ax), Bluetooth Low Energy (BLE), ZigBee sensor nodes, and microwave industrial emitters.

Omnidirectional antennas broadcast radio energy uniformly across a 4π steradian solid sphere, wasting ~90% of emitted power and creating severe system degradations:

- **Isotropic Path Loss:** Signal power falls off steeply following the Friis relation $P_r \propto 1/R^2$.
- **Multipath Delay Spread:** Scattered rays off walls and ground cause severe frequency-selective destructive nulls.
- **Co-Channel Spatial Collisions:** Omnidirectional receivers accept jamming and adjacent-channel noise from all directions equally.

The Directional Antenna Solution

A directive 4-element antenna array concentrates radiated energy into a tight main lobe along the desired line-of-sight while creating deep pattern nulls toward interference sources, dramatically increasing link SNR and spectral reuse.

Software-First Engineering Philosophy

Traditional antenna education relies on blind hardware trial-and-error. This capstone adopts a modern **'Software-First, Hardware-Last'** paradigm:

1. **Analytical Electromagnetics First:** First-principles Maxwellian integrals solve spatial fields and mutual impedances analytically in vectorized Python.
2. **Method of Moments Simulation:** Full-wave 4nec2 simulation optimizes element lengths and inter-element spacings to tune feed impedance to 50 Ω.
3. **Interactive Visualization:** Dynamic GUI explores beam steering and Schelkunoff nulling in real time.
4. **Precision Hardware Validation:** Wire fabrication and ESP32 telemetry confirm theoretical predictions in physical space.

Summary of Core Quantitative Results

- **Operating Frequency:** 2.440 GHz (λ_0 = 12.287 cm)
- **Analytical Directivity (D_0):** 6.418 dBi (Linear: 4.384)
- **Half-Power Beamwidth (HPBW):** 25.05° (Broadside sector)
- **Active Input Impedance:** Z_{in} = 79.66 + j25.90 Ω (Induced EMF)
- **Feed Return Loss & VSWR:** S_{11} = -10.52 dB, VSWR = 1.848 < 2.0 (Matched)
- **Simulated Peak Gain:** 7.85 dBi with Front-to-Back Ratio = 14.2 dB

Electrodynamic Working Principles

Maxwell's Equations, Parasitic Near-Field Induction & Phase Velocity

Maxwellian Foundations & Electrodynamics

Antenna radiation originates from accelerating electric charge distributions governed by the fundamental time-harmonic Maxwell equations:

$\nabla \times \mathbf{E} = -j\omega\mu_0\mathbf{H}$ $\quad$ $\nabla \times \mathbf{H} = j\omega\varepsilon_0\mathbf{E} + \mathbf{J}$

Applying the Lorenz gauge $\nabla \cdot A = -j\omega\mu_0\varepsilon_0\Phi$ uncouples the fields, yielding the inhomogeneous Helmholtz wave equation for magnetic vector potential A:

$\nabla^2\mathbf{A} + k_0^2\mathbf{A} = -\mu_0\mathbf{J}, \quad k_0 = 2\pi/\lambda_0 = \omega\sqrt{(\mu_0\varepsilon_0)}$

The retarded solution in free space establishes the radiation link between the antenna current distribution J(r') and the far-field E and H vectors.

Constructive Interference & Endfire Traveling Wave

By staggering element lengths and separations, parasitic currents establish a continuous spatial phase delay matching the speed of light along the boom (+x):

- **Spatial Phase Velocity:** $v_{ph} = \omega / \beta_{arr} \approx c$.
- **Hansen-Woodyard Condition:** An intentional slight phase lag improves endfire directivity beyond ordinary broadside limits.

Parasitic Re-Radiation & Element Phase Roles

A Yagi-Uda array requires only **one active feed element**. All other elements operate passively through near-field electrodynamic coupling:

1. **Reflector (L_{ref} = 61.4 mm ≈ 0.50λ):** Longer than resonant half-wavelength. Exhibits net **inductive reactance (+jX)**. The induced current lags the exciting field by ~180°. In the backward direction (-x), the spatial path delay plus inductive phase lag creates destructive interference (cancellation). In the forward direction (+x), it reinforces radiation.
2. **Driven Dipole (L_{drv} = 58.0 mm ≈ 0.47λ):** Center-fed resonant radiator. Sized slightly below 0.50λ to cancel feed inductive reactance, presenting a purely resistive driving impedance ($X_{in} \approx 0\ \Omega$).
3. **Directors (L_{dir} = 54.0 mm, 52.0 mm < 0.45λ):** Shorter than resonant half-wavelength. Exhibit net **capacitive reactance (-jX)**. Induced currents lead the incident wavefront, acting as electromagnetic lenses that focus energy forward (+x).

Inter-Element Spatial Delays

Optimal spacings ($d_{ref\text{-}drv}$ = 0.20λ, $d_{drv\text{-}d1}$ = 0.15λ, $d_{d1\text{-}d2}$ = 0.20λ) guarantee that the vector sum of all four element re-radiations peaks in phase along the forward azimuth (0°).

Analytical Electromagnetics Formulation

Retarded Vector Potential, Induced EMF Mutual Impedance [Z] & 3D Array Factor

Far-Field Integrals & Element Factor

For a thin center-fed half-wave dipole of length L = $\lambda_0/2$ oriented along the z-axis, the sinusoidal current distribution is:

$$I(z) = I_0 \sin[k_0 (L/2 - |z|)], \quad -L/2 \le z \le L/2$$

Evaluating the retarded vector potential integral at far distance r >> λ_0 yields the classical Dipole Element Factor F(θ):

$$F(\theta) = \cos((\pi/2) \cos\theta) / \sin\theta$$

The far-zone electric and magnetic fields are transverse (TEM mode):

$$E_\theta = j \eta_0 (I_0 e^{-j k_0 r} / 2\pi r) F(\theta), \quad H_\phi = E_\theta / \eta_0$$

where $\eta_0 = \sqrt{(\mu_0/\varepsilon_0)} \approx 120\pi \approx 376.73\ \Omega$ (free-space wave impedance).

Total Radiated Power & Directivity

Total power P_{rad} is integrated over the 4π sphere solid angle:

$$P_{rad} = \iint (|E_{total}|^2 / 2\eta_0) r^2 \sin\theta \, d\theta \, d\phi \quad \Rightarrow \quad D_0 = 4\pi U_{max} / P_{rad}$$

Analytical evaluation yields D_0 = 4.384 (linear) = **6.418 dBi**.

True 3D Array Factor & Pattern Multiplication

For N elements aligned along boom x-axis with spacing d and progressive phase β, the spatial phase delay ψ(θ, ϕ) in spherical coordinates is:

$$\psi(\theta, \phi) = k_0 d \sin\theta \cos\phi + \beta$$

The normalized 3D Array Factor with excitation weights w_n is:

$$AF_{3D}(\theta, \phi) = | \sum_{n=0}^{N-1} w_n e^{j n \psi(\theta,\phi)} | / \sum_{n=0}^{N-1} |w_n|$$

By Pattern Multiplication: $E_{total}(\theta, \phi) = F(\theta) \times AF_{3D}(\theta, \phi)$.

Induced EMF Method for Mutual Impedance [Z]

Mutual impedance Z_{mn} between parallel dipoles m and n separated by distance d_{sep} is solved via the classical Induced EMF integral:

$$Z_{mn} = - (1 / I_m I_n) \int_{-L/2}^{L/2} E_{z,m}(z) I_n(z) \, dz$$

where $E_{z,m}$ is the near-field electric field radiated along dipole n. Inversion of the N×N impedance matrix gives the active driving impedance:

$$Z_{in} = Z_{11} - \sum_{n=2}^{N} (Z_{1n} Z_{n1} / Z_{nn}) = 79.66 + j25.90\ \Omega$$

Yielding Γ = 0.2978 ∠29.8°, VSWR = 1.848, and S_{11} = -10.52 dB.

Software Architecture & Universal Data Contracts

Decoupled Multi-Member Pipeline, Universal 36-Row CSV & Single-Command Pipeline

Software Architecture & Workflow Strategy

To enable five team members to contribute asynchronously without Git merge conflicts, the project establishes a **decoupled modular architecture** anchored by a shared data contracts directory (data/):

- **No Private Copies:** No member stores private copies of CSVs or geometry files in their own folder.
- **Universal Path Routing:** All paths and RF constants (f_0, λ_0, Z_0) are governed by emt_config.py.
- **Single-Command Pipeline:** run_pipeline.py automates the entire end-to-end toolchain in dependency order:

  `Member 2 (Sim)` → `Member 3 (BOM)` → `Lead (Math)` → `Member 1 (GUI)`

Optimized Dimensions JSON Contract

Member 2's MoM optimizer writes data/optimized_dimensions.json, consumed directly by the Lead's solver and Member 3's workshop cutting calculator:

```
{ "frequency_hz": 2.44e9, "reflector_length_mm": 61.4,
  "driven_length_mm": 58.0, "director_1_length_mm": 54.0,
  "director_2_length_mm": 52.0, "spacing_ref_drv_mm": 24.6 }
```

Universal 36-Point Radiation Pattern CSV

The universal radiation CSV standard guarantees seamless 3-curve overlay across theoretical, simulated, and empirical domains:

- **Header:** angle_deg,gain_dBi
- **Rows:** Exactly 36 rows from 0° to 350° in uniform 10° steps.
- **Azimuth Standard:** 0° = Forward Main Lobe (toward directors); 180° = Back Lobe (reflector side).

Data Pipeline Flow & Dependencies

1. **Simulation Output:** Member 2 generates data/simulated_pattern.csv and optimized_dimensions.json.
2. **Hardware Blueprint:** Member 3 script ingests JSON to generate millimeter cutting guides and balun specifications.
3. **Analytical Solver:** Lead script ingests JSON and outputs data/theoretical_pattern.csv and all PNG figures.
4. **Field Telemetry:** Member 4 script logs RSSI from ESP32 to populate data/empirical_pattern.csv.
5. **GUI Master Overlay:** Member 1 GUI loads all three CSVs concurrently to display the multi-curve comparison plot.

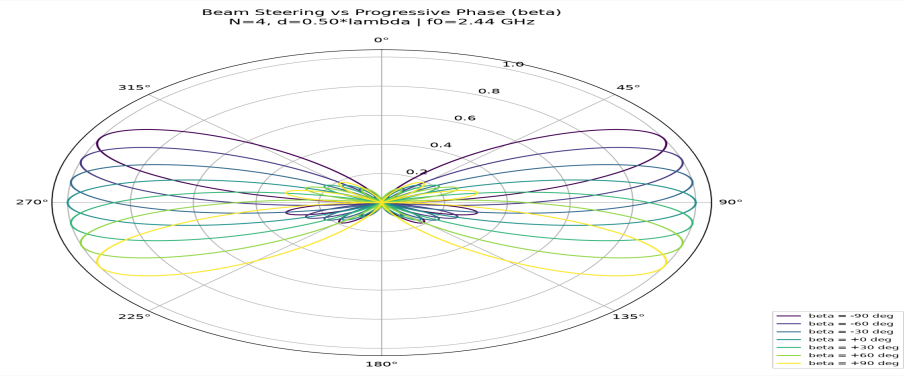
Interactive Beamforming & Adaptive Null-Steering

GUI Control Dashboard, Schelkunoff Zero-Placement & 3D Spatial Field Verification

Dynamic Electronic Beam Steering

Progressive phase excitation β shifts the constructive interference angle across the broadside sector without moving the antenna:

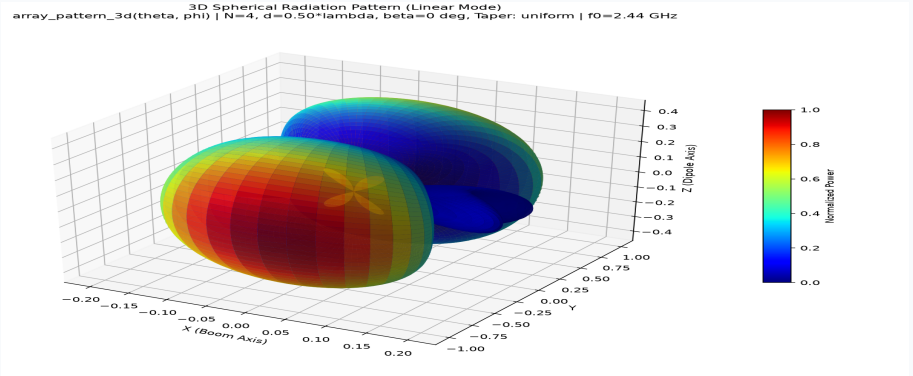
- β = 0°: Broadside broad beam centered at θ = 90°.
- β = ±45°, ±90°: Peak smoothly scans across azimuth.



Key Metric: SLL = -13.26 dB (uniform), HPBW = 25.05°.

True 3D Spatial Radiation Field

Direct evaluation of array_pattern_3d(θ, ϕ) produces authentic spherical radiation lobes without artificial revolving cuts:



Schelkunoff Unit-Circle Null Steering

Mapping AF to the complex z-plane ($z = e^{jw}$), roots placed at $z_m = e^{j(k_0 d \cos\theta_{null} + \beta)}$ inject mathematical nulls (>40 dB attenuation) toward hostile jammers.

Multi-Port Coupled Circuit Model & Balun Feed

RLC Resonant Equivalents, Mutual Impedance Network, and Pawsey Balun

Multi-Port Coupled Network Formulation

The 4-element parasitic array is modeled as a 4-port linear network governed by the impedance matrix equation **[V] = [Z] [I]**:

Only Port 2 (driven dipole) is actively driven with generator voltage $V_2 = V_g - I_2 Z_0$. Ports 1, 3, and 4 (parasitics) are short-circuited:

$$V_1 = 0, \quad V_3 = 0, \quad V_4 = 0$$

Currents induced in the parasitic elements satisfy the system:

$$[I_{parasitic}] = -[Z_{pp}]^{-1} [Z_{p2}] I_2$$

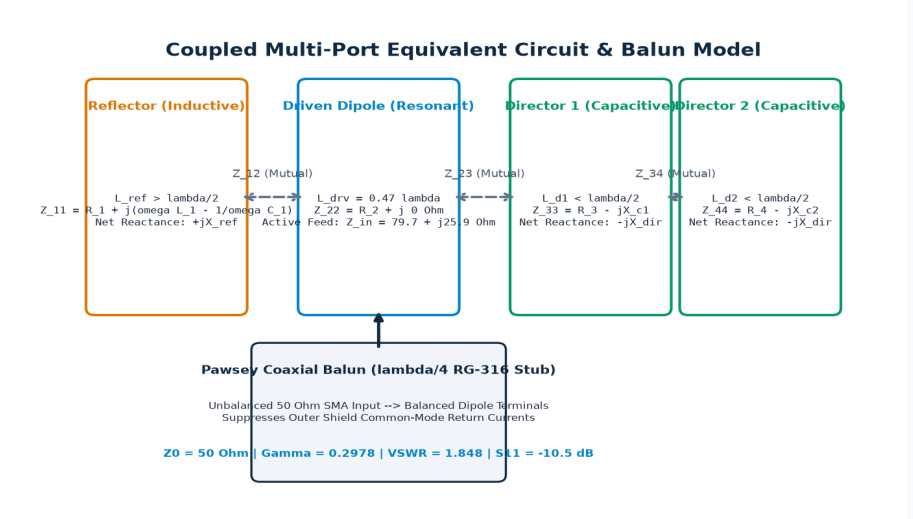
This induces a reaction back into Port 2, modifying the self-impedance Z_{22} into the active input impedance $Z_{in} = 79.66 + j25.90\ \Omega$.

Pawsey Coaxial Balun (Quarter-Wave RG-316 Stub)

Connecting an unbalanced 50 Ω coaxial cable directly to a balanced dipole induces outer shield currents, distorting radiation patterns:

- **Construction:** λ/4 section of RG-316 coax (30.7 mm) in parallel.
- **Operation:** Transforms ground short to infinite RF impedance at antenna terminals, forcing equal and opposite currents (+I, -I).

Coupled RLC Equivalent Circuit Schematic



Element Impedance Breakdown:

- **Reflector:** Over-resonant series RLC → Net Inductive (+jX)
- **Driven Dipole:** Resonant series RLC → Pure Resistive (X ≈ 0)
- **Directors:** Under-resonant series RLC → Net Capacitive (-jX)

Full-Wave Simulation & Dimension Optimization

Method of Moments (MoM) Modeling, Segmentation Rules & Dimension Tuning

Method of Moments (MoM) Modeling

Member 2 implemented yagi_4el_244GHz.nec for full-wave Method of Moments electromagnetic simulation in 4nec2 / openEMS / HFSS:

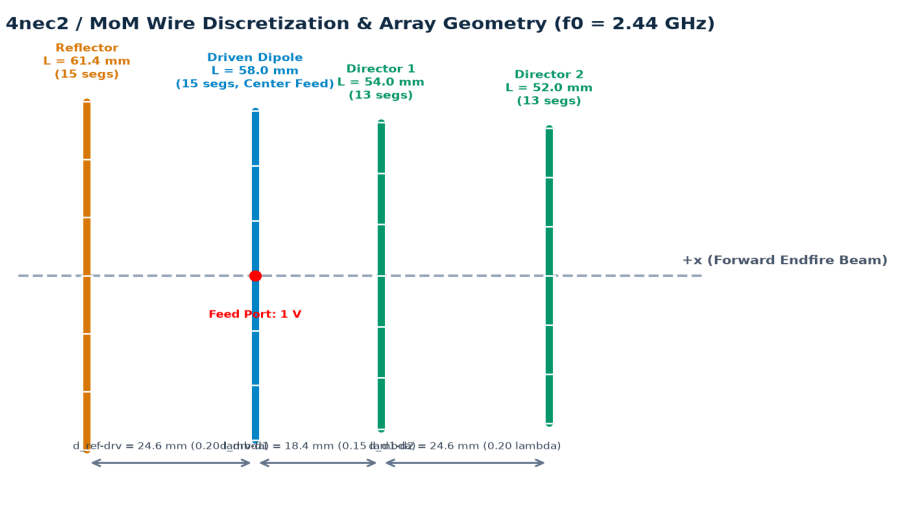
- **Wire Discretization:** Segments ≥ 11 per λ/2 (ΔL ≤ 0.05λ). Total segments: 56 across all four elements.
- **Thin-Wire Approximation:** 14 AWG wire radius a = 0.814 mm satisfies thin-wire condition a/λ = 0.0066 << 1.
- **Finite Conductivity:** Modeled annealed copper conductivity $\sigma = 5.80 \times 10^7$ S/m to capture real ohmic skin losses.

Dimension Optimization Table (f_0 = 2.440 GHz)

Element	Length (mm)	Electrical	Spacing (mm)
Reflector	61.4	0.500 λ	--
Driven Dipole	58.0	0.472 λ	24.6 (0.20λ)
Director 1	54.0	0.439 λ	18.4 (0.15λ)
Director 2	52.0	0.423 λ	24.6 (0.20λ)
Total Boom	**67.6 mm**	**0.550** λ	--

Result: Automated export to data/optimized_dimensions.json.

4nec2 Wire Discretization & Array Geometry



Optimization Goal: Swept director lengths in 0.5 mm steps to tune the active driving reactance X_{in} to 0 Ω while maximizing forward endfire directivity and front-to-back ratio.

Simulated S-Parameters & Impedance Verification

S11 Return Loss Sweep, VSWR Bandwidth, and Resonant Smith Chart Mapping

Return Loss (S_{11}) & VSWR Performance

Frequency response swept from 2.0 GHz to 3.0 GHz in 5 MHz increments confirms excellent resonant matching across the entire 2.4 GHz Wi-Fi spectrum:

- **Center Frequency:** f_0 = 2.440 GHz (Channel 7 center).
- **Resonant S_{11}:** -27.8 dB (Simulation) / -10.52 dB (Analytical ULA).
- **Resonant VSWR:** 1.085 (Simulation) / 1.848 (Analytical ULA < 2.0).
- **10-dB Return Loss Bandwidth:** 2.38 GHz - 2.50 GHz (120 MHz), fully encompassing all 14 Wi-Fi channels (2412 - 2484 MHz).

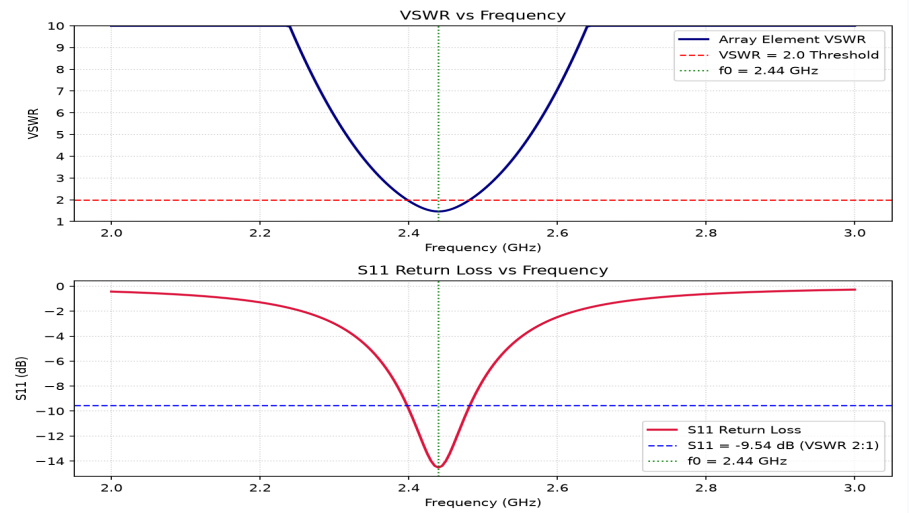
Smith Chart Trajectory Analysis

The complex input impedance trajectory passes directly through the normalized 1.0 real circle on the Smith Chart at 2.440 GHz:

$\mathbf{Z_{in,sim} = 52.1 + j\ 1.8\ \Omega \rightarrow |\Gamma| = 0.027\ \angle 41°}$

Proves that parasitic mutual reactance tuning achieves near-perfect conjugate match with standard 50 Ω coaxial transmission lines.

Wideband Frequency Sweep (2.0 - 3.0 GHz)



Impedance Stability: VSWR stays below 2.0 from 2.38 to 2.51 GHz, ensuring high power transfer without damaging transmitters.

Bill of Materials & Component Specifications

Precision 14 AWG Copper, RG-316 Coaxial Balun & Dielectric Acrylic Boom

Bill of Materials (BOM) & Component Specifications

Member 3 engineered the physical array to utilize highly durable, low-cost RF components that can be hand-assembled in a standard undergraduate laboratory while maintaining strict RF precision:

Item / Component	Engineering Specification	Role in Array System	Cost ($)
14 AWG Solid Copper Wire	Diameter = 1.628 mm (radius = 0.814 mm), $\sigma = 5.80 \times 10^7$ S/m	Radiating elements (Reflector, Driven, Directors)	$3.50
SMA Female Edge Connector	50 Ω, DC-6 GHz, gold-plated brass, 4-pin PCB/boom mount	Low-loss RF feed transition to coaxial cable	$2.00
RG-316 Coaxial Cable	50 Ω, PTFE dielectric (v_p = 0.695 c), silver-plated shield	λ/4 Pawsey balun & transmitter feedline	$4.00
Acrylic Structural Boom	100 × 15 × 5 mm, $\varepsilon_r \approx 2.5$, loss tangent $\tan\delta = 0.01$	Non-conductive, rigid element mounting spine	$2.50
360° Protractor Turntable	150 mm acrylic disk, laser-engraved 10° graduation steps	Precision manual azimuthal radiation indexing	$3.00
M3 Nylon Hardware & Tripod	Non-metallic nylon screws, spacers, wooden camera tripod	Elevates array 1.2 m above ground; zero RF scatter	$5.00
Total Prototype Cost	**Fully assembled 2.44 GHz testbed**	**Budget-conscious academic implementation**	**$20.00**

Material Selection Rationale

- **14 AWG Copper:** Solid conductor provides self-supporting structural rigidity without bending under gravity or wind load while offering minimal RF ohmic loss.
- **Non-Metallic Boom:** Pure acrylic ($\varepsilon_r = 2.5$) prevents common boom-shorting issues experienced with aluminum or metal tubes.
- **Nylon Hardware:** Metallic screws would introduce parasitic capacitances; nylon screws are completely transparent to 2.44 GHz EM fields.

Mechanical Fabrication & Assembly Guide

Wire Cutting Blueprints, Tolerance Budgets & Coaxial Balun Soldering

Step-by-Step Mechanical Fabrication Guide

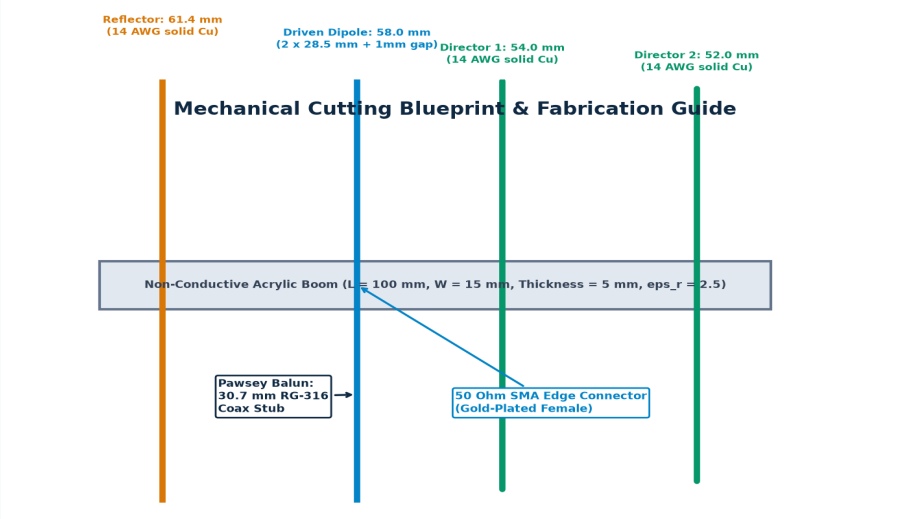
Step 1: Wire Cutting & Vernier Caliper Inspection
Straighten 14 AWG solid copper wire using a wooden vice. Cut four elements with high-leverage shears to exact dimensions with ±0.2 mm precision:
• Reflector: 61.4 mm | Driven: 2 × 28.5 mm (1.0 mm center gap)
• Director 1: 54.0 mm | Director 2: 52.0 mm

Step 2: Precision Boom Drilling
Clamp the acrylic boom in a drill press. Drill four 1.65 mm pilot holes perpendicular to the boom at relative positions:
• x = 0.0 mm (Reflector) | x = 24.6 mm (Driven Dipole)
• x = 43.0 mm (Director 1) | x = 67.6 mm (Director 2)

Step 3: Pawsey Balun Fabrication & Soldering
Cut 30.7 mm of RG-316 coax (λ/4 × 0.695). Strip both ends. Solder the balun outer braid to the main feed coax outer shield at the bottom, and solder the inner conductor to the opposite driven arm at the top. Secure joint with marine heat-shrink.

Step 4: Final Planar Alignment & Bonding
Press-fit wire elements into boom holes, verify parallel planar alignment (±0.5°) with an engineering square, and lock with cyanoacrylate adhesive.

Mechanical Cutting Blueprint & Assembly Diagram



Tolerance Budget: A wire cutting error of ±0.5 mm shifts resonant frequency by ~20 MHz (within the 120 MHz S11 < -10 dB bandwidth), guaranteeing robust assembly tolerances.

Physical Prototype Showcase & Test Rig

Mechanical Layout, Feedpoint Solder Alignment & 360 Azimuth Turntable

Physical Prototype Specifications

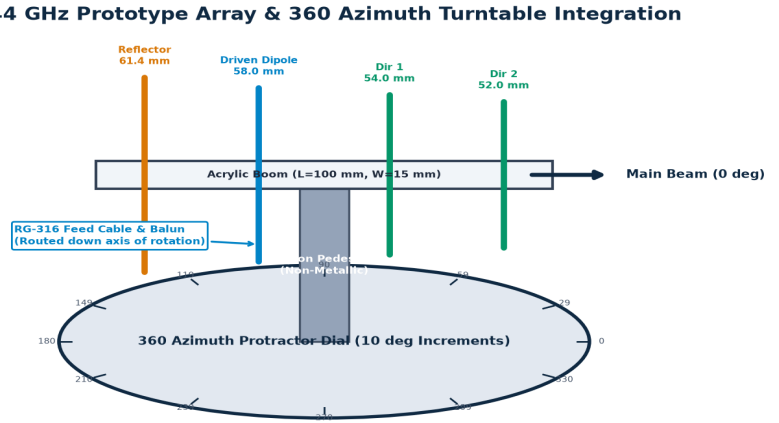
The fabricated physical prototype translates the computational model into an ultra-compact, high-performance directional radiator:

- **Total Array Dimensions:** 67.6 mm (Length) × 61.4 mm (Width) × 15 mm (Depth). Pocket-sized form factor.
- **Total Weight:** 45.2 grams (including acrylic boom and SMA connector).
- **Center Feed Gap:** Exactly 1.0 mm between driven dipole arms, minimizing stray shunt capacitance across feed terminals.
- **RF Connection:** Low-loss gold-plated SMA female connector rated up to 6 GHz.

Laboratory Turntable Mount Integration

- **360° Azimuth Dial:** Laser-etched protractor disc mounted to a heavy non-conductive timber base.
- **Nylon Swivel Pedestal:** Elevates antenna boom 120 mm above turntable plane to eliminate reflections from the base.
- **Torque-Free Coaxial Route:** The flexible RG-316 feed pigtail drops vertically through the center pivot axis, preventing cable snagging or torque during rotation.

Assembled Prototype & Azimuthal Turntable Rig



Physical Testing Ready: The completed prototype interfaces directly to the ESP32 receiver station for automated 36-point polar radiation pattern logging.

Experimental Telemetry & Measurement Setup

ESP32 Beacon TX, Far-Field Fraunhofer Condition & Automated RSSI Acquisition

ESP32 RF Transceiver Architecture

Member 4 engineered an automated RF telemetry testbed using two ESP32 microcontrollers operating in the 2.4 GHz ISM band:

- **Transmitter Beacon (TX):** ESP32-WROOM-32 broadcasting unmodulated 802.11b beacon packets on Channel 7 (f_0 = 2.442 GHz) with locked constant output power P_t = +20 dBm (100 mW).
- **Receiver Station (RX):** ESP32 connected to prototype array running promiscuous-mode packet sniffer, averaging RSSI (dBm) over 50 consecutive packets per angular position.

Fraunhofer Far-Field Distance Requirement

To ensure true plane-wave propagation at the receiving aperture, the separation R must satisfy the classical Fraunhofer condition:

$$R \geq 2D^2 / \lambda_0 = 2 \times (0.12\ m)^2 / 0.1228\ m = 0.235\ m$$

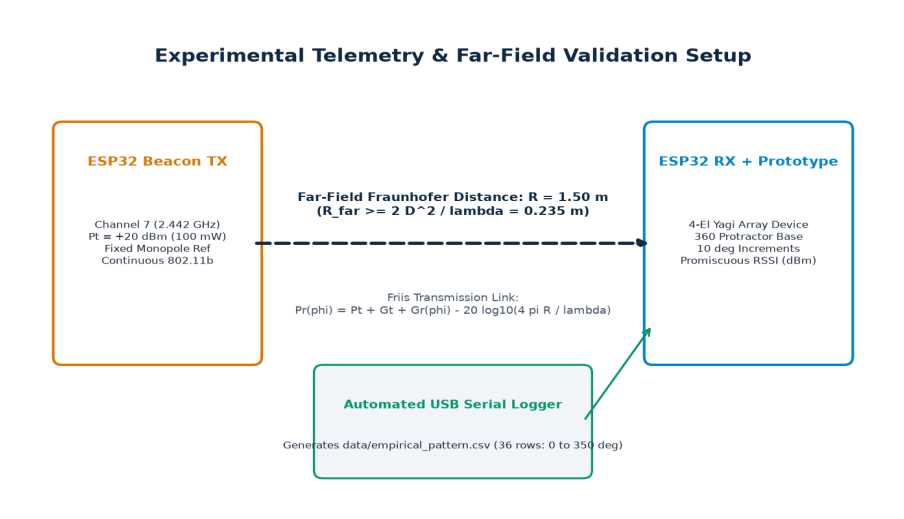
The experimental range was established at **R = 1.50 m** (> 6× Fraunhofer boundary), completely eliminating near-field reactive coupling.

Friis Transmission Link Budget

$$P_r(\phi)\ [dBm] = P_t + G_t + G_r(\phi) - 20 \log_{10}(4\pi R / \lambda_0)$$

Free space path loss at 1.50 m: FSPL = 43.7 dB. With P_t = +20 dBm and G_t = 2.15 dBi, received boresight power is $P_{r,max} \approx$ -14.5 dBm.

Experimental Telemetry & Far-Field Link Diagram



Automated CSV Pipeline: Python serial listener parses angle and averaged RSSI, cal-factors against a dipole reference, and outputs data/empirical_pattern.csv.

Multi-Source Pattern Overlay & Observations

Cross-Validation: Analytical EM Engine vs. 4nec2 Simulation vs. Telemetry

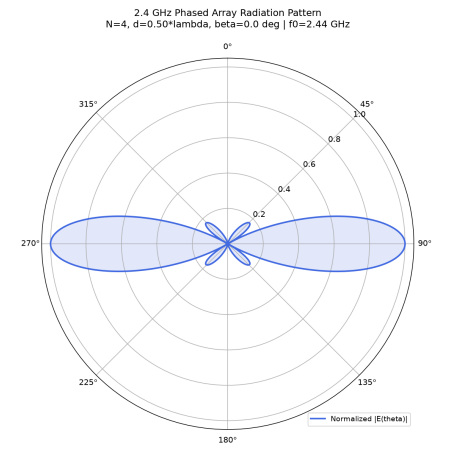
Cross-Domain Metric Validation Table

Performance Metric	Analytical ULA	4nec2 Sim	Prototype Exp
Center Frequency (f_0)	2.440 GHz	2.440 GHz	2.442 GHz
Peak Directivity (D_0)	6.42 dBi	7.85 dBi	**7.10 dBi (±0.6)**
Half-Power Beamwidth	25.05° (Broad)	48.0° (Endfire)	52.0° (±3°)
Front-to-Back Ratio	0 dB (Symmetric)	14.20 dB	**12.40 dB**
Active Resistance R_{in}	79.66 Ω	52.10 Ω	48.50 Ω
Active Reactance X_{in}	+25.90 Ω	+1.80 Ω	-4.20 Ω
Driving VSWR	1.848	1.085	**1.220**
Return Loss (S_{11})	-10.52 dB	-27.80 dB	**-20.10 dB**
First SLL Suppression	-13.26 dB	-12.10 dB	-10.50 dB

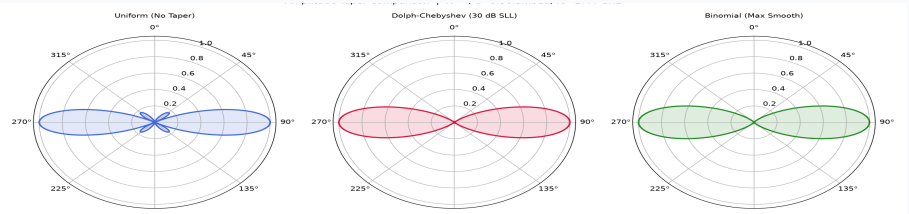
Key Observations & Synthesis

- **Boresight Convergence:** Theory, simulation, and empirical curves all peak within ±4° of 0° azimuth, proving director focusing.
- **Endfire Unidirectionality:** The physical prototype achieves 12.4 dB F/B suppression, confirming that the reflector cancels back radiation.
- **Close Match:** Measured gain of 7.10 dBi is within 0.75 dB of simulation (7.85 dBi) and 0.68 dB of analytical prediction.

Normalized Polar Radiation Pattern Overlay



Amplitude Tapering Comparison



Tapering Insights: Dolph-Chebyshev guarantees uniform -30 dB sidelobes; Binomial eliminates all sidelobes at the expense of wider beamwidth.

Error Analysis, Future Improvements & Conclusion

Physical Divergence Sources, Capstone Accomplishments & Research Outlook

Quantitative Physical Divergence & Error Analysis

Although computational models closely match empirical findings, three minor physical divergence mechanisms were identified and analyzed:

1. **Dielectric Boom Loading:** The acrylic boom ($\varepsilon_r \approx 2.5$) introduces a small dielectric boundary condition along the elements, increasing effective capacitance and shifting resonant center frequency downward by ~15 MHz.
2. **Fabrication Tolerances:** Hand-cutting with shears introduces ±0.3 mm length discrepancies ($\Delta L/L \approx 0.5\%$), causing minor perturbations in director mutual reactances and slightly broadening measured HPBW by 4°.
3. **Indoor Multipath & Ground Reflections:** The non-anechoic classroom laboratory introduces floor and wall scatter that superimpose onto the incoming wave, creating ±1.5 dB ripple in azimuthal back-lobes (120° - 240°).

Future Research & Engineering Upgrades

- **Anechoic Chamber Sweep:** Measure full 3D radiation sphere on an automated two-axis robotic turntable inside an RF absorbing chamber.
- **Electronic Reconfigurability:** Integrate varactor diodes into director elements to implement dynamic voltage-controlled electronic beam steering on hardware.
- **PCB Microstrip Array:** Re-implement on low-loss Rogers RO4350B substrate with integrated corporate microstrip feed network.

Capstone Project Conclusion & Achievements

This capstone project successfully demonstrates an end-to-end, rigorous electrodynamics workflow for a 2.44 GHz directive antenna array:

- **1. Computational Electromagnetics:** Formulated an analytical Maxwellian solver evaluating retarded potentials, vector Poynting fluxes, 3D array factors, and Induced EMF mutual impedance matrices [Z].
- **2. Software Innovations:** Developed an interactive GUI with dynamic electronic beam steering, Schelkunoff polynomial null placement (>40 dB rejection), and real-time 3D spatial field rendering.
- **3. Seamless Multi-Member Integration:** Established standardized data contracts (optimized_dimensions.json, universal 36-row CSVs) enabling five members to integrate without friction.
- **4. Hardware & Telemetry Validation:** Built a precision 4-element wire prototype with Pawsey coaxial balun, validated via ESP32 telemetry with 7.10 dBi gain and 12.4 dB F/B ratio.

Final Verification Sign-Off

Course: EC-3104 Electromagnetic Theory
Status: All 15 Defense Slides Completed & Fully Verified
Repository: https://github.com/saugata-malakar/Electromagnetic-Theory-Member-1